

MAANAV ANAND KUMAR

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Education

- Stony Brook University, *Master of Science in Computer Science*** **Aug. 2024 – May 2026**
– **GPA:** 3.7/4* New York, USA
– **Relevant Coursework:** Analysis of Algorithms, Operating Systems, Human Computer Interaction, Wireless Networks, Mobile Robotics, Hybrid Systems, AI for Robotics, Computer Graphics
- Chennai Institute of Technology, *Bachelor of Engineering in Computer Science*** **Aug. 2020 – May 2024**
– **GPA:** 8.91/10 Chennai, India
– **Relevant Coursework:** Distributed Systems, Data Structures and Algorithms, Computer Networks, Software Engineering, Object Oriented Programming, Database Management Systems, Cloud Computing, Machine Learning

Experience

- New York Wealth Planning Group, Software Engineer Intern** **New York, USA | May 2025 – Aug. 2025**
 - Deployed full-stack web applications using React.js, Node.js, and MongoDB on AWS EC2, architected to handle over 10K concurrent sessions. Implemented CI/CD pipelines using GitHub Actions and Docker, ensuring seamless scalability.
 - Designed and implemented RESTful APIs secured via Auth0, leveraging optimized database queries, Redis caching, and connection pooling layers that reduced average response time by 22% and supported over 150K monthly transactions.
- CGI Information Systems, SDE Apprentice (ML Systems)** **Chennai, India | May 2024 – July 2024**
 - Developed and optimized machine learning models using Python, TensorFlow, and scikit-learn, integrating them into containerized services with Docker and Kubernetes to enhance automation and scalability of agile predictive analytics modules.
 - Refactored core components to improve model execution speed, optimize memory allocation, and streamline data I/O, achieving a ~15% reduction in latency and enhancing the reliability and throughput of distributed ML deployments.
- Timescan Logistics, Software Developer Intern** **Chennai, India | Jun. 2023 – Dec. 2023**
 - Engineered automation scripts and data integration to synchronize multi-branch logistics operations, reducing manual workload by 35% and improving turnaround time for trans-continental shipment processing, contributing to measurable business growth.
 - Maintained enterprise-grade web applications for global freight and shipment management using Flask, MySQL, and RESTful APIs, improving data access efficiency and system reliability of internal tools.

Projects

- Web Based Telemetry and Control Dashboard** **Interacting Robotic Systems Lab, New York**
 - Built a modular web-based telemetry dashboard enabling real-time visualization and remote control of IoT and robotic devices using a microservice architecture with Flask, WebSockets, and ROSBridge, ensuring reliable data exchange across distributed nodes.
 - Designed backend pipelines for concurrent sensor streams, achieving sub-200 ms message latency through asynchronous I/O, optimized serialization, and Redis-backed message caching. Deployed on AWS EC2 instances (Linux) using Docker.
 - Integrated fine tuned LLM components to enable natural language command parsing and automated system diagnostics, improving operator usability and reducing manual configuration overhead.
- Software Framework for ML Training and Evaluation Platform** **Stony Brook University, New York**
 - Developed a scalable deep reinforcement learning (DRL) framework in Python using TensorFlow and PyTorch to simulate and evaluate autonomous navigation policies for high-speed, collision-free driving in virtual F1TENTH track environments.
 - Implemented a robust API layer and configuration management system to automate experiment setup, parameter tuning, and performance tracking across model variants, supporting seamless integration with data visualization dashboards.
- Gesture Based Navigation Control Application** **Stony Brook University, New York**
 - Engineered a real-time vision-based cursor control application using OpenCV and Tobii SDK, enabling intuitive human-computer interaction through head tracking and gesture recognition with over 95% tracking accuracy across diverse lighting conditions.
 - Optimized the end-to-end video processing pipeline using multithreading, Kalman filtering, and asynchronous neural inference, reducing per-frame latency to under 45 ms and improving responsiveness on standard computing platforms.

Skills

Programming Languages Python, C++, C#, Java, JavaScript, SQL, HTML
Frameworks & Libraries Flask, Django, React.js, Node.js, TensorFlow, PyTorch, ROS/ROS2, OpenCV
Cloud & Databases GCP, AWS, MySQL, PostgreSQL, MongoDB, Redis
Developer Tools Docker, Kubernetes, Git, GitHub Actions, JIRA
Software Engineering REST APIs, Microservices, CI/CD, Object-Oriented Programming, Unit Testing

Publications

Investigating the Prediction of Breast Cancer Diagnosis by Use of Support Vector Machines
International Journal of Healthcare Information Systems and Informatics (IJHISI) Jun. 2023

Leadership Roles

Computer Science Senator, Stony Brook University **Sep. 2024 – May 2025**
Section Leader - Mentor for Stanford Code in Place **May 2024 – Jun. 2024**